

# Snowflake CLI – Homework / Play Tasks

## 1. Basic CLI Usage

### 1.1 Check the Snowflake CLI version:

**Snow --version**

```
C:\Users\DELL>snow --version  
Snowflake CLI version: 3.13.0
```

### 1.2 Test your connection to Snowflake using the CLI:

- - Snow connection add

-- Snow Connection Test -- connection hari-con

```
C:\Users\DELL>snow connection test --connection hari_con
```

| key             | value                                  |
|-----------------|----------------------------------------|
| Connection name | hari_con                               |
| Status          | OK                                     |
| Host            | YJFVGCR-SF46636.snowflakecomputing.com |
| Account         | YJFVGCR-SF46636                        |
| User            | HARIHARAN12                            |
| Role            | ACCOUNTADMIN                           |
| Database        | not set                                |
| Warehouse       | COMPUTE_WH                             |

## 2. Running Simple SQL

### 2.1 Run a SQL command to show all databases.

```
snow sql -c hari_con -q "show Databases;"
```

| created_on                       | name                  | is_default | is_current | origin                                           | owner        | comment                                                               | options | retention_time | kind              | owner_role_type | object_visibility |
|----------------------------------|-----------------------|------------|------------|--------------------------------------------------|--------------|-----------------------------------------------------------------------|---------|----------------|-------------------|-----------------|-------------------|
| 2025-12-02 06:09:37.790000-08:00 | SNOWFLAKE             | N          | N          | SNOWFLAKE_ACCOUNT_USAGE                          |              |                                                                       |         | 0              | APPLICATION       |                 | None              |
| 2025-12-02 06:09:45.924000-08:00 | SNOWFLAKE_LEARNING_DB | N          | N          |                                                  | ACCOUNTADMIN | Created by Snowflake during account provisioning                      |         | 1              | STANDARD          | ROLE            | None              |
| 2025-12-02 06:09:43.728000-08:00 | SNOWFLAKE_SAMPLE_DATA | N          | N          | SFSALESSHAREDSFCSAMPLESACENTRALINDIA.SAMPLE_DATA | ACCOUNTADMIN | Preloaded TPCB Data provided by Snowflake during account provisioning |         | 0              | IMPORTED DATABASE | ROLE            | None              |
| 2025-12-09 00:26:21.721000-08:00 | TRAINING              | N          | N          |                                                  | ACCOUNTADMIN |                                                                       |         | 1              | STANDARD          | ROLE            | None              |
| 2025-12-03 21:31:05.725000-08:00 | USERSHARIHARAN12      | N          | N          |                                                  |              |                                                                       |         | 1              | PERSONAL DATABASE |                 | None              |

```
snow sql -c hari_con -q "show warehouses;"
```

| id         | Project Overview |          |         |      |                   |                   |            |         |        |            | Development & Testing |              |             |           |              |           |       |                            |                            |                            | Deployment & Monitoring |          |                           |                  |        |         |           |           |      |                |                 |                     |                         |               |
|------------|------------------|----------|---------|------|-------------------|-------------------|------------|---------|--------|------------|-----------------------|--------------|-------------|-----------|--------------|-----------|-------|----------------------------|----------------------------|----------------------------|-------------------------|----------|---------------------------|------------------|--------|---------|-----------|-----------|------|----------------|-----------------|---------------------|-------------------------|---------------|
|            | name             | state    | type    | size | min_cluster_count | max_cluster_count | start_date | running | queued | is_default | is_current            | auto_suspend | auto_resume | available | provisioning | quiescing | other | created_on                 | resume_on                  | updated_on                 | owner                   | comment  | enable_query_acceleration | resource_monitor | active | pending | failed    | suspended | uid  | scaling_policy | owner_role_type | resource_constraint | warehouse_created_limit | target_status |
| COMPUTE_WH | SUPPLEMENTAL     | STANDARD | X-Small | 1    | 1                 | 0                 | 0          | 0       | Y      | Y          | 600                   | true         |             |           |              |           |       | 2025-01-22 06:17:16.750000 | 2025-01-22 08:24:21.000000 | 2025-01-22 08:24:21.000000 | ADMIN                   | UNTAMING | false                     | 0                | 0      | 1       | 216962724 | STANDARD  | ROLE | STANDARD       | None            | None                | None                    |               |

```
- -snow sql -c hari con -f "C:\Users\DELL\Documents\classignment.sql"
```

```

C:\Users\DELL\Documents>snow sql -c hari_con -f "C:\Users\DELL\Documents\cliassignment.sql"
USE DATABASE cli_db;
+-----+
| status |
+-----+
| Statement executed successfully. |
+-----+

USE SCHEMA cli_schema;
+-----+
| status |
+-----+
| Statement executed successfully. |
+-----+

CREATE OR REPLACE TABLE cli_employee (
  emp_id INT,
  emp_name VARCHAR(50),
  designation VARCHAR(50)
);
+-----+
| status |
+-----+
| Table CLI_EMPLOYEE successfully created. |
+-----+

INSERT INTO cli_employee (emp_id, emp_name, designation) VALUES
(1, 'Hari', 'HR'),
(2, 'Kavin', 'Engineer'),
(3, 'Giri', 'Analyst');
+-----+
| number of rows inserted |
+-----+
| 3 |
+-----+

```

```
SELECT * FROM cli_employee;
```

```

+-----+
| EMP_ID | EMP_NAME | DESIGNATION |
+-----+
| 1      | Hari     | HR          |
| 2      | Kavin    | Engineer    |
| 3      | Giri     | Analyst     |
+-----+

```

### 3. Creating Objects

#### 3.1 Create a database using the CLI.

Snow sql -c hari\_con -q "CREATE OR REPLACE cli\_db;"

```
C:\Users\DELL\Documents>snow sql -c hari_con -q "create or replace database cli_db;"
create or replace database cli_db;
+-----+
| status |
+-----+
| Database CLI_DB successfully created. |
+-----+
```

### 3.2 Create a schema inside that database

Snow sql -c hari\_con -q "create or replace cli\_db.cli\_schema;"

```
C:\Users\DELL\Documents>snow sql -c hari_con -q "create or replace schema cli_db.cli_schema;"
create or replace schema cli_db.cli_schema;
+-----+
| status |
+-----+
| Schema CLI_SCHEMA successfully created. |
+-----+
```

### 3.3 Create a table using a SQL file and run it through the CLI.

cd C:\Users\DELL\Documents - - - First we change the directory in our file location

snow sql -c hari\_con -f cliassignment.sql - - - And we put file name

```
C:\Users\DELL\Documents>snow sql -c hari_con -f cliassignment.sql
CREATE OR REPLACE DATABASE cli_demo_db;
+-----+
| status |
+-----+
| Database CLI_DEMO_DB successfully created. |
+-----+

USE DATABASE cli_demo_db;
+-----+
| status |
+-----+
| Statement executed successfully. |
+-----+

CREATE OR REPLACE SCHEMA cli_demo_schema;
+-----+
| status |
+-----+
| Schema CLI_DEMO_SCHEMA successfully created. |
+-----+

USE SCHEMA cli_demo_schema;
+-----+
| status |
+-----+
| Statement executed successfully. |
+-----+
```

## 4. Working With Stages

### 4.1 Upload a small CSV file to an internal stage using the CLI.

snow sql -c hari\_con -q "create or replace stage cli\_db.cli\_schema.cli\_stage;"

```
C:\Users\DELL\Documents>snow sql -c hari_con -q "create or replace stage cli_db.cli_schema.cli_stage;"
create or replace stage cli_db.cli_schema.cli_stage;
+-----+
| status |
+-----+
| Stage area CLI_STAGE successfully created. |
+-----+
```

```
C:\Users\DELL\Documents>snow sql -c hari_con -q "PUT file:///C:/Users/DELL/DOCUMENTS/cli.csv @cli_db.cli_schema.cli_stage;"
PUT file:///C:/Users/DELL/DOCUMENTS/cli.csv @cli_db.cli_schema.cli_stage;
+-----+
| source | target | source_size | target_size | source_compression | target_compression | status | message |
+-----+-----+-----+-----+-----+-----+-----+-----+
| cli.csv | cli.csv.gz | 432 | 304 | NONE | GZIP | UPLOADED | 
```

#### 4.2 List the files inside the stage.

snowsql -c hari\_con -q "LIST @cli\_db.cli\_schema.cli\_stage;"

```
C:\Users\DELL\Documents>
C:\Users\DELL\Documents>snow sql -c hari_con -q "LIST @cli_db.cli_schema.cli_stage;"
LIST @cli_db.cli_schema.cli_stage;
+-----+
| name | size | md5 | last_modified |
+-----+-----+-----+-----+
| cli_stage/cli.csv.gz | 304 | 7fe8f2b129398a96b99cd46ce014e7bc | Wed, 10 Dec 2025 16:52:12 GMT |
+-----+
```

#### 4.3 Download the uploaded file back to your computer.

Snow sql -c hari\_con -q "Get @cli\_db.cli\_schema.cli-stage/cli.csv  
file:///c:/Users/DELL/Downloads/;"

```
C:\Users\DELL\Documents>snow sql -c hari_con -q "GET @cli_db.cli_schema.cli_stage/cli.csv file:///C:/Users/DELL/Downloads/;"
GET @cli_db.cli_schema.cli_stage/cli.csv file:///C:/Users/DELL/Downloads/;
+-----+
| file      | size | status | message |
+-----+
| cli.csv.gz | 295  | UPLOADED |         |
+-----+
```

## 5. Loading and Unloading Data

### 5.1 Load CSV data into a table using COPY INTO via the CLI.

Snow sql -c hari\_con -q "create or replace table cli\_db.cli\_schema.cli\_emp(emp\_id int,emp\_name varchar(50),designation varchar(50));

```
C:\Users\DELL>snow sql -c hari_con -q "CREATE OR REPLACE TABLE cli_db.cli_schema.cli_emp (emp_id int,emp_name varchar(50),designation varchar(50));
CREATE OR REPLACE TABLE cli_db.cli_schema.cli_emp (emp_id int,emp_name varchar(50),designation varchar(50));
+-----+
| status |
+-----+
| Table CLI_EMP successfully created. |
+-----+
```

Snow sql -c hari\_con -q "PUT file:///c:/users/dell/documents/clidata.csv @cli\_db.cli\_schema.cli\_stage;"

```
C:\Users\DELL>snow sql -c hari_con -q "PUT file:///c:/users/dell/documents/clidata.csv @cli_db.cli_schema.cli_stage;"
PUT file:///c:/users/dell/documents/clidata.csv @cli_db.cli_schema.cli_stage;
+-----+
| source      | target      | source_size | target_size | source_compression | target_compression | status | message |
+-----+
| clidata.csv | clidata.csv.gz | 210         | 176         | NONE               | GZIP               | UPLOADED |         |
+-----+
```

Snow sql -c hari\_con -q "copy into cli\_db.cli\_schema.cli\_emp from @cli\_db.cli\_schema.cli\_stage File\_Format=(Type = csv Skip\_Header=1)on\_error=continue;"

```
C:\Users\DELL>snow sql -c hari_con -q "copy into cli_db.cli_schema.cli_emp from @cli_db.cli_schema.cli_stage File Format =(Type = csv Skip_Header=1)on_error=continue;"
copy into cli_db.cli_schema.cli_emp from @cli_db.cli_schema.cli_stage File Format =(Type = csv Skip_Header=1)on_error=continue;
```

| file                     | status | rows_parsed | rows_loaded | error_limit | errors_seen | first_error | first_error_line | first_error_character | first_error_column_name |
|--------------------------|--------|-------------|-------------|-------------|-------------|-------------|------------------|-----------------------|-------------------------|
| cli_stage/clidata.csv.gz | LOADED | 10          | 10          | 10          | 0           | None        | None             | None                  | None                    |

## 5.2 Unload data from a table into a stage.

Snow\_sql -c hari\_con -q "copy into @cli\_db.cli\_stage from cli\_db.cli\_schema.cli\_emp  
File\_Format=(TYPE=CSV) OVERWRITE=TRUE;"

```
C:\Users\DELL>snow sql -c hari_con -q "COPY INTO @cli_db.cli_schema.cli_stage FROM cli_db.cli_schema.cli_emp FILE_FORMAT=(TYPE=CSV) OVERWRITE=TRUE;"
COPY INTO @cli_db.cli_schema.cli_stage FROM cli_db.cli_schema.cli_emp FILE_FORMAT=(TYPE=CSV) OVERWRITE=TRUE;
```

| rows_unloaded | input_bytes | output_bytes |
|---------------|-------------|--------------|
| 10            | 173         | 139          |

## 6. Deploying a Snowflake Project

Initialize a Snowflake project using snow init.

Deploy the project using snow deploy.

## 7. Query Output Practice

### 7.1 Run a SELECT query and save the results to a CSV file using the CLI.

Snow sql -c hari\_con -q "SELECT \* FROM cli\_db.cli\_schema.cli\_emp;" -o output\_format=csv -o  
output\_file="C:\Users\DELL\Downloads\mydata.csv"

```
C:\Users\DELL>
C:\Users\DELL>snow sql -c hari_con -q "select*from cli_db.cli_schema.cli_emp;"
select*from cli_db.cli_schema.cli_emp;
```

| EMP_ID | EMP_NAME | DESIGNATION |
|--------|----------|-------------|
| 101    | Ravi     | Manager     |
| 102    | Kumar    | Developer   |
| 103    | Sara     | Tester      |
| 104    | Mani     | HR          |
| 105    | Arun     | Support     |
| 106    | Deepa    | Developer   |
| 107    | Vijay    | Manager     |
| 108    | Priya    | Tester      |
| 109    | Karthik  | Developer   |
| 110    | Anitha   | HR          |

Snow sql -c hari\_con -q "select\*from cli\_db.cli\_schema.cli\_emp;">  
c:\users\DELL\Downloads\emp.csv

```
C:\Users\DELL>snow sql -c hari_con -q "SELECT * FROM cli_db.cli_schema.cli_emp;" > C:\Users\DELL\Downloads\emp.csv
```

## 7.2 Try different output formats such as pretty, CSV, and JSON.

C:\Users\DELL>snow sql -c hari\_con -q "SELECT \* FROM cli\_db.cli\_schema.cli\_emp;" >  
C:\Users\DELL\Downloads\emp.json

```
C:\Users\DELL>snow sql -c hari_con -q "SELECT * FROM cli_db.cli_schema.cli_emp;" > C:\Users\DELL\Downloads\emp.json
```

## 8. Playing With Context

### 8.1 Change the warehouse while running a query.

Snow sql -c hari\_con -q "use warehouse cli\_wh;select\*from cli\_db.cli\_schema.cli\_emp;"

```
C:\Users\DELL>Snow sql -c hari_con -q "use warehouse cli_wh;select*from cli_db.cli_schema.cli_emp;"
use warehouse cli_wh;
```

```
+-----+
| status |
+-----+
| Statement executed successfully. |
+-----+
```

```
select*from cli_db.cli_schema.cli_emp;
```

```
+-----+
| EMP_ID | EMP_NAME | DESIGNATION |
+-----+
| 101    | Ravi     | Manager     |
| 102    | Kumar    | Developer    |
| 103    | Sara     | Tester       |
| 104    | Mani     | HR           |
| 105    | Arun     | Support      |
| 106    | Deepa    | Developer    |
| 107    | Vijay    | Manager      |
| 108    | Priya    | Tester       |
| 109    | Karthik  | Developer    |
| 110    | Anitha   | HR           |
+-----+
```

## 8.2 Change the database and schema using CLI options.

```
C:\Users\DELL>snow sql -c hari_con --database cli_db --schema cli_schema -q "SELECT * FROM cli_emp;"
SELECT * FROM cli_emp;
```

```
+-----+
| EMP_ID | EMP_NAME | DESIGNATION |
+-----+
| 101    | Ravi     | Manager     |
| 102    | Kumar    | Developer    |
| 103    | Sara     | Tester       |
| 104    | Mani     | HR           |
| 105    | Arun     | Support      |
| 106    | Deepa    | Developer    |
| 107    | Vijay    | Manager      |
| 108    | Priya    | Tester       |
| 109    | Karthik  | Developer    |
| 110    | Anitha   | HR           |
+-----+
```

## 9. Troubleshooting and Help

### 9.1 Use the help command to list all available CLI commands.

```

C:\Users\DELL>snow stage --help

Usage: -c stage [OPTIONS] COMMAND [ARGS]...

Manages stages.

Options
--help -h      Show this message and exit.

Commands
copy           Copies all files from source path to target directory. This works for uploading to and downloading files from the stage, and copying between named
               stages.
create         Creates a named stage if it does not already exist.
describe       Provides description of stage.
drop           Drops stage with given name.
execute        Execute immediate all files from the stage path. Files can be filtered with a glob-like pattern, e.g. @stage/*.sql, @stage/dev/*. Only files with .sql
               extension will be executed.
list           Lists all available stages.
list-files     Lists the stage contents.
remove         Removes a file from a stage.

```

## 9.2 Run a command in verbose mode to view detailed logs.

- Verbose mode shows you everything that's happening behind the scenes when a command runs.
- It gives extra details like steps, logs, and hidden messages that you normally don't see.
- We use it to easily understand problems and figure out why something didn't work.

Snow sql -c hari\_con --verbose -q "SELECT CURRENT\_DATE();"

```

C:\Users\DELL>snow sql -c hari_con --verbose -q "SELECT CURRENT_DATE();"
SELECT CURRENT_DATE();
+-----+
| CURRENT_DATE() |
+-----+
| 2025-12-11     |
+-----+

```